P.1 (Simon B.1)

for bcc we can choose the basis :

$$a_1 = (1,0,0)\,a$$
$$a_2 = (0,1,0)\,a$$
$$a_3 = (\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2})\,a$$

I

&

or as it turns out to be useful :

$$a_3 - a_1 = (-\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2})\,a$$
$$a_3 - a_2 = (\tfrac{1}{2},-\tfrac{1}{2},\tfrac{1}{2})\,a$$
$$a_1 + a_2 - a_3 = (\tfrac{1}{2},\tfrac{1}{2},-\tfrac{1}{2})\,a$$

I'

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for fcc we have the basis :

$$a_1 = (0,\tfrac{1}{2},\tfrac{1}{2})\,a$$
$$a_2 = (\tfrac{1}{2},0,\tfrac{1}{2})\,a$$
$$a_3 = (\tfrac{1}{2},\tfrac{1}{2},0)\,a$$

II

&

and again the useful basis would be :

$$a_1 + a_2 - a_3 = (0,0,1)\,a$$
$$a_3 - a_1 = (\tfrac{1}{2},0,-\tfrac{1}{2})\,a$$
$$a_3 - a_2 = (0,\tfrac{1}{2},-\tfrac{1}{2})\,a$$

II'

Now calculate the reciprocal lattice basis using the formula

$$\vec{b}_i = 2\pi \frac{\vec{a}_j \times \vec{a}_k}{a_i \cdot (\vec{a}_j \times \vec{a}_k)}$$

for fcc we get : $b_3 = \frac{4\pi}{a}(\tfrac{1}{2}\ \tfrac{1}{2}\ -\tfrac{1}{2})$

which is bcc I' with $b_1 = \frac{4\pi}{a}(-\tfrac{1}{2}\ \tfrac{1}{2}\ \tfrac{1}{2})$

$\boxed{4\pi/a}$ as the lattice constant $b_2 = \frac{4\pi}{a}(\tfrac{1}{2}\ -\tfrac{1}{2}\ \tfrac{1}{2})$

And for bcc we get:

which is a fcc basis II'

for lattice constant $\frac{4\pi}{a}$

$$b_3 = \frac{4\pi}{a}(0, 0, 1)$$
$$b_1 = \frac{4\pi}{a}(\tfrac{1}{2}, 0, -\tfrac{1}{2})$$
$$b_2 = \frac{4\pi}{a}(0, \tfrac{1}{2}, -\tfrac{1}{2})$$

The basis for face-centered orthorombic (fco) would be:

$$\vec{c}_1 = (0, \tfrac{a_2}{2}, \tfrac{a_3}{2})$$
$$\vec{c}_2 = (\tfrac{a_1}{2}, 0, \tfrac{a_3}{2})$$
$$c_3 = (\tfrac{a_1}{2}, \tfrac{a_2}{2}, 0)$$

using our formula: $b_i = 2\pi \dfrac{c_j \times c_k}{c_i \cdot (c_j \times c_k)}$

we get:

$$\boxed{\begin{aligned} b_1 &= 2\pi\left(-\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}\right) \\ b_2 &= 2\pi\left(\frac{1}{a_1}, -\frac{1}{a_2}, \frac{1}{a_3}\right) \\ b_3 &= 2\pi\left(\frac{1}{a_1}, \frac{1}{a_2}, -\frac{1}{a_3}\right) \end{aligned}}$$

P.2 (Simon 13.6)

a) Cubic lattice:

$a_1 = a(0,0,1)$ $\qquad b_1 = \frac{2\pi}{a}(0,0,1)$ $\qquad b_i = 2\pi \frac{a_j \times a_k}{a_i(a_j \times a_k)}$

$a_2 = a(0,1,0) \rightarrow b_2 = \frac{2\pi}{a}(0,1,0)$ $\qquad$ used

$a_3 = a(1,0,0)$ $\qquad b_3 = \frac{2\pi}{a}(1,0,0)$

So $\vec{b}$'s also form an orthogonal basis. The Fist BZ would be a cube of length $\frac{2\pi}{a}$ centered at the origin.

In other words any vector like $k = \frac{2\pi}{a}(x,y,z)$ where $|x|$ & $|y|$ & $|z| < \frac{1}{2}$ is in the FBZ.

for arbitrary vector like $\vec{k} = \frac{2\pi}{a}(u,v,w)$ we can write it as $k = \frac{2\pi}{a}(m+x, n+y, p+z)$ where $m,n,p \in \mathbb{Z}$ and $x, y, z \in [-\frac{1}{2}, \frac{1}{2}]$ then the vector $k' = \frac{2\pi}{a}(x,y,z)$ would k's equivalent in FBZ.

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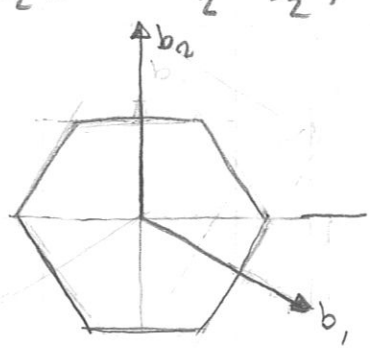
b) triangular lattice:

$a_1 = a(1,0)$ $\qquad b_1 = \frac{4\pi}{\sqrt{3}a}(\frac{\sqrt{3}}{2}, -\frac{1}{2})$ $\hat{z}$ is used as *

$a_2 = a(\frac{1}{2}, \frac{\sqrt{3}}{2}) \rightarrow b_2 = \frac{4\pi}{\sqrt{3}a}(0,1)$ the fictitious third basis



The FBZ can be found by drawing the perpendicular bisector of $\pm\vec{b}_2$, $\pm\vec{b}_1$ and $\pm(\vec{b}_1 + \vec{b}_2)$

which is a hexagone with $\frac{4\pi}{\sqrt{3}a} \times \frac{1}{2} \times \frac{2}{\sqrt{3}} = \frac{4\pi}{3a}$

as its side length.

Then to find k', equivalent to arbitrary k in FBZ,
First we have to find α_1 & α_2 such that:
$\vec{k} = \alpha_1 \vec{b_1} + \alpha_2 \vec{b_2}$. To do that we write

$k \cdot b_1 = \alpha_1 |b_1|^2 + \alpha_2 b_1 \cdot b_2$ with $|b_1|^2 = |b_2|^2 = \left(\frac{4\pi}{\sqrt{3}a}\right)^2 = \gamma$

$k \cdot b_2 = \alpha_1 b_1 \cdot b_2 + \alpha_2 |b_2|^2$ and $b_1 \cdot b_2 = -\frac{1}{2}\gamma$

we have $\left.\begin{array}{l} k \cdot b_1 = \gamma(\alpha_1 - \frac{1}{2}\alpha_2) \\ k \cdot b_2 = \gamma(\alpha_2 - \frac{1}{2}\alpha_1) \end{array}\right\}$ from which we can solve for α_1 & α_2 and get

$$\alpha_1 = \frac{2}{3\gamma}(2k \cdot b_1 + k \cdot b_2)$$

$$\alpha_2 = \frac{2}{3\gamma}(2k \cdot b_2 + k \cdot b_1)$$

Now if we subtract some integer part from α_1 & α_2 we still get an equivalent $\vec{k}'$ vector to the original one.
Therefore, to find the equivalent k' in 1BZ one need subtract some integer from α_1 & α_2 such that the remaining is in $[-\frac{1}{2}, \frac{1}{2}]$ or formally speaking:

$k' = \alpha_1' b_1 + \alpha_2' b_2$ where $\alpha_1 = m + \alpha_1'$ & $\alpha_2 = n + \alpha_2'$
m & $n \in \mathbb{Z}$ and $|\alpha_1'|$ & $|\alpha_2'| < \frac{1}{2}$